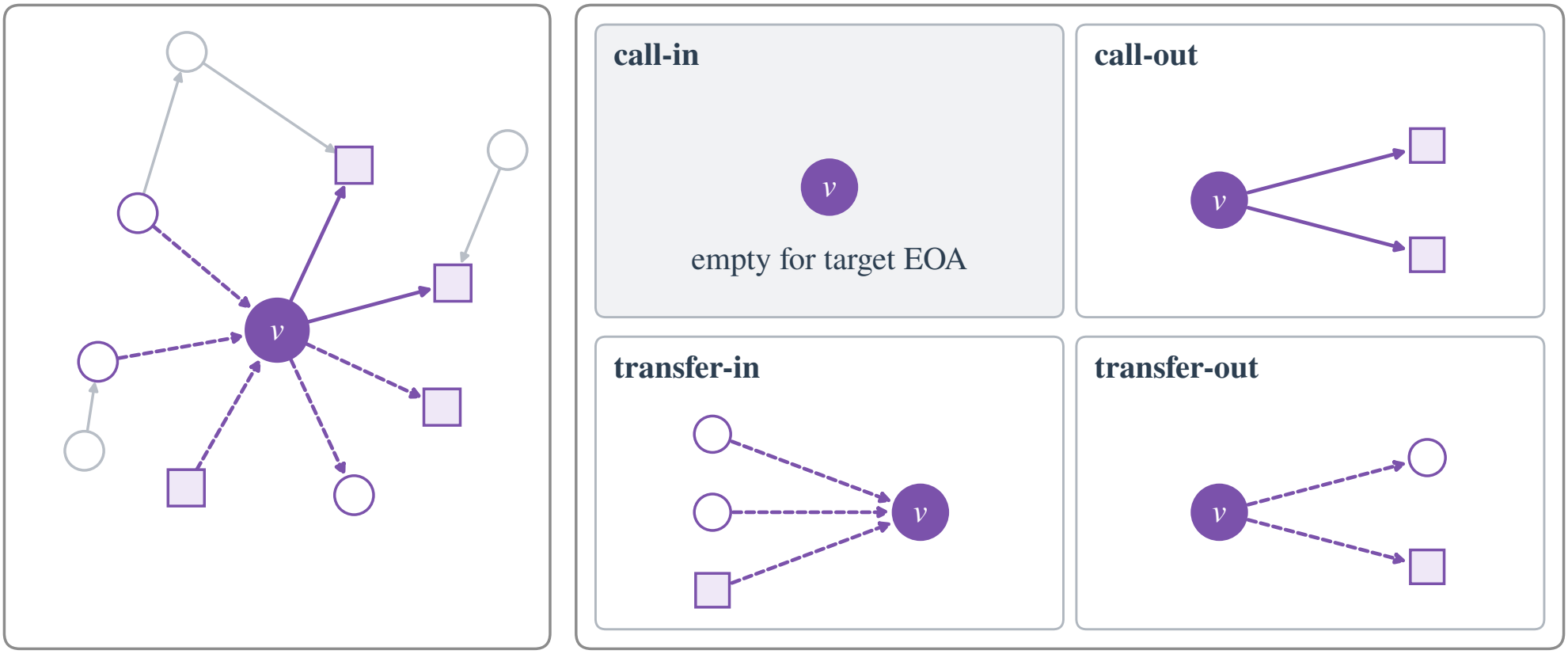

(a) Target EOA neighbourhood **(b) Relation–direction decomposition $E_{r,d}(v)$**

for each group $E_{r,d}(v)$:

$g_{r,d,n_e}(v) = |E_{r,d}(v)|$

$g_{r,d,\Sigma a_k}(v) = \Sigma_{e \in E_{r,d}(v)} a_k(e),\ k \in \{0,1\}$

$u_{r,d,s}(v) = \log(1 + g_{r,d,s}(v))$

$\mathbf{u}(v) \in \mathbb{R}^{12}$

call-in			call-out			transfer-in			transfer-out		
n_e	Σa_0	Σa_1	n_e	Σa_0	Σa_1	n_e	Σa_0	Σa_1	n_e	Σa_0	Σa_1
0	0	0									

(c) Aggregation into the 12-dimensional UFE vector

v target EOA ○ EOA □ CA → call ⇢ transfer → non-incident edge (not aggregated)

UFE entry structurally zero for target EOA a_0, a_1: released numerical edge attributes